ECON1010 Introductory Microeconomics

Tutorial 4 | Demand and Supply II

This Week in ECON1010

CML 2 is open and due **next week** 4.00pm March 31

CLEAR-JE 2 is due **next week** 4.00pm April 02

Marking for CLEAR-JE 1 is underway

Assessment

85%	7
75%	6
65%	5
50%	4
47%	3
30%	2
0%	1

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP…

1. **Ed Discussion Board** (Blackboard)
2. **Consultation** (8.00am Mon, 39-125A)
3. **Staff email** econ1010@uq.edu.au

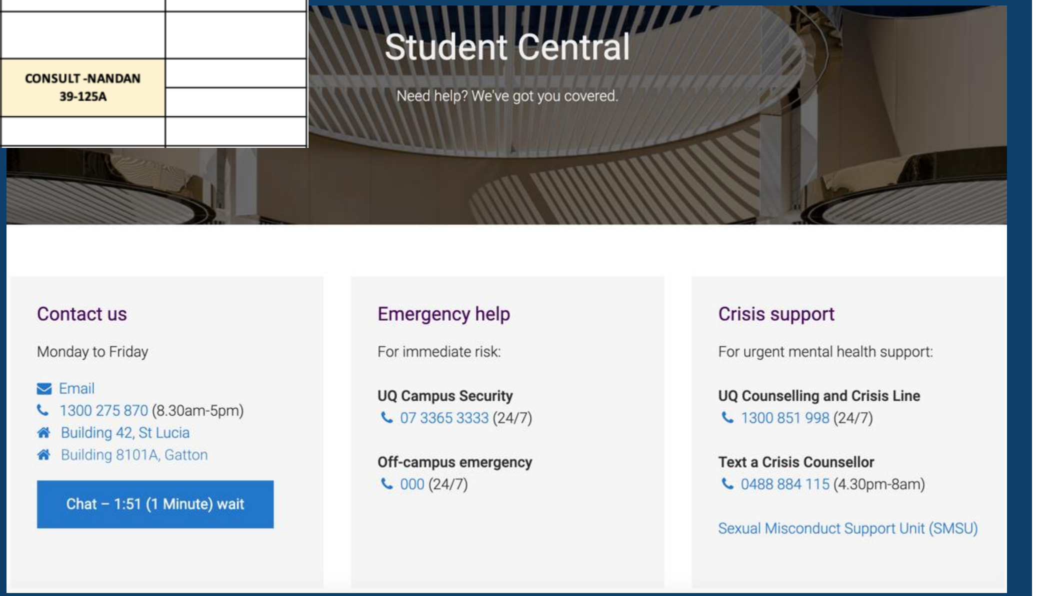
I REALLY need HELP…

- https://my.uq.edu.au/contact/student-central

ECON1010 Consult Timetable

Semester 1 2026

Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT -NANDAN 39-125A	
16:30-17:00					
17:00-17:30					



Demand and Supply II

Question 1a

a) Consider the following questions:

i) What are two factors that can cause a shift in Demand?

ii) What are two factors that can cause a shift in Supply?

iii) In your own words, what are substitute goods?

iv) What is an example complement good for an iPhone?

v) How do future price expectations affect current prices? Why?

Discuss with your table group, choose one person to share

Question 1a

i) What are two factors that can cause a shift in Demand?

1. **Change in consumer taste (or preference) – due to marketing/trends**
2. **Change in population**
3. **Change in expectations of future price rises**
4. **The change in demand for complement/substitute goods**
5. **Change in disposable income**

ii) What are two factors that can cause a shift in Supply?

1. **Change in number of suppliers in the market**
2. **Change in expectations of future selling price**
3. **Change in level of input costs**
4. **Change production to a new, substitute product**
5. **Changes in technology**
6. **Changes in government taxes on a product**

Demand Side

Non-price factors

Income
Number of buyers
Substitute demand
Expectations
Complement demand
Tastes

Supply Side

Non-price factors

Alternative goods
Unexpected events
Number of sellers
Technology

Cost of making goods
Joint supply

Question 1a

iii) In your own words, what are substitute goods?

Replacement products or services. Example: iPhone 6 v iPhone 6S – a person will use one or the other, but not both together. When the iPhone 6S was released, sales of the iPhone 6 dropped significantly.

v) What is an example complement good for an iPhone?

Mobile data, phone cases, headphones, speakers, applications, chargers, apple watch, electricity.

vi) How do future price expectations affect current prices? Why?

If prices are going to increase in the future and you are going to buy the good in the future, it is better value to buy it today. As demand increases (because people are buying more of it today), the price will rise towards the expected future price – and vice-versa.

Question 1b

b) Draw a graph of supply and demand (with axes labels) for each situation below. Show the impact on the curve(s), and explain your answer:

i) Demand: Increase in population (with supply fixed)

ii) Supply: Technological improvement (with demand fixed)

iii) Supply: More expensive input costs (with demand fixed)

Discuss with your table group, choose one person to share

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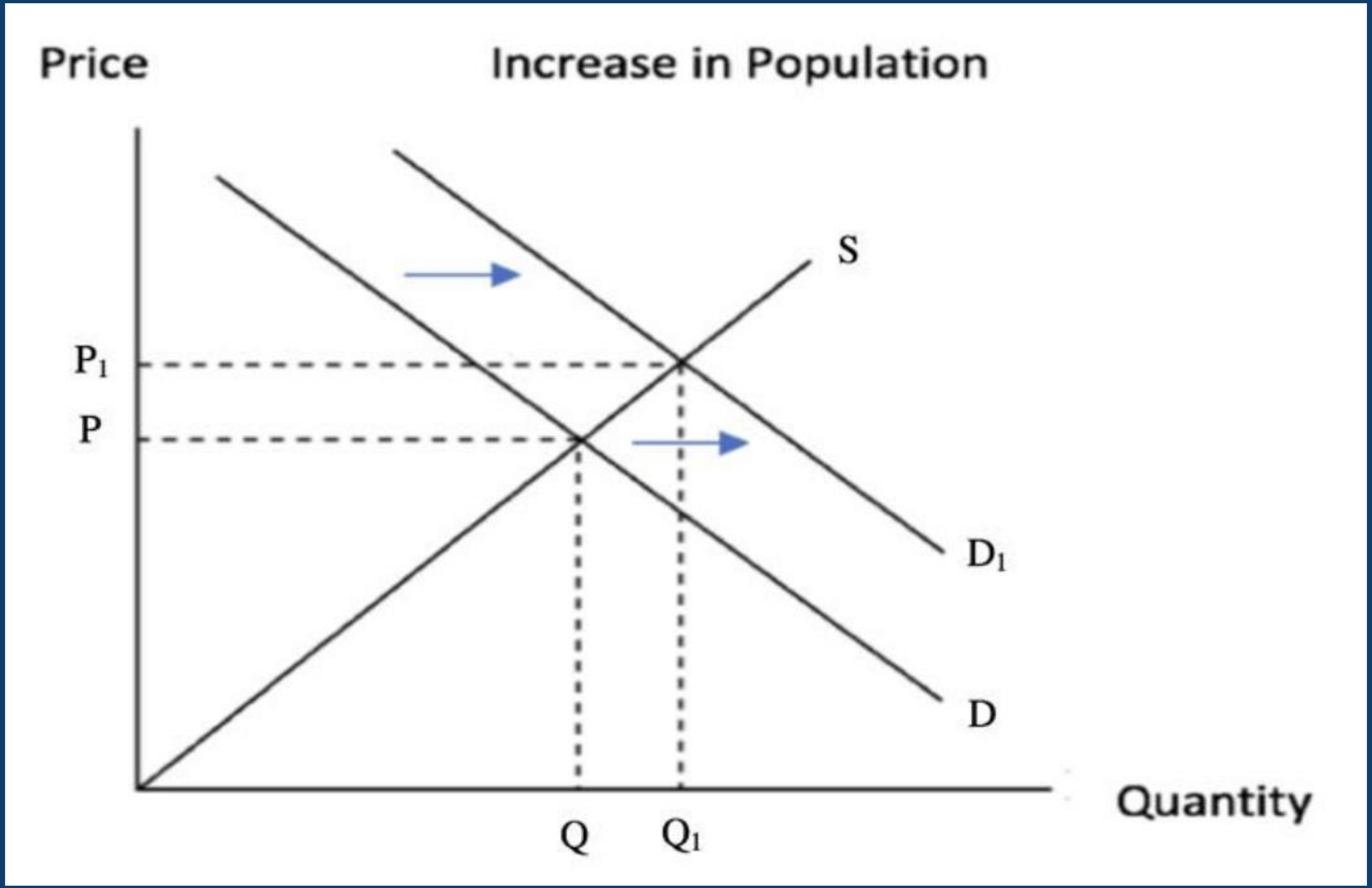
Question 1b

i) Demand: Increase in population (with supply fixed)

An increase in populations results in more participants in the market and therefore my quantity demanded for the same price level. Therefore, the demand curve shifts right. Quantity increase to Q_1 and price increase to P_1.

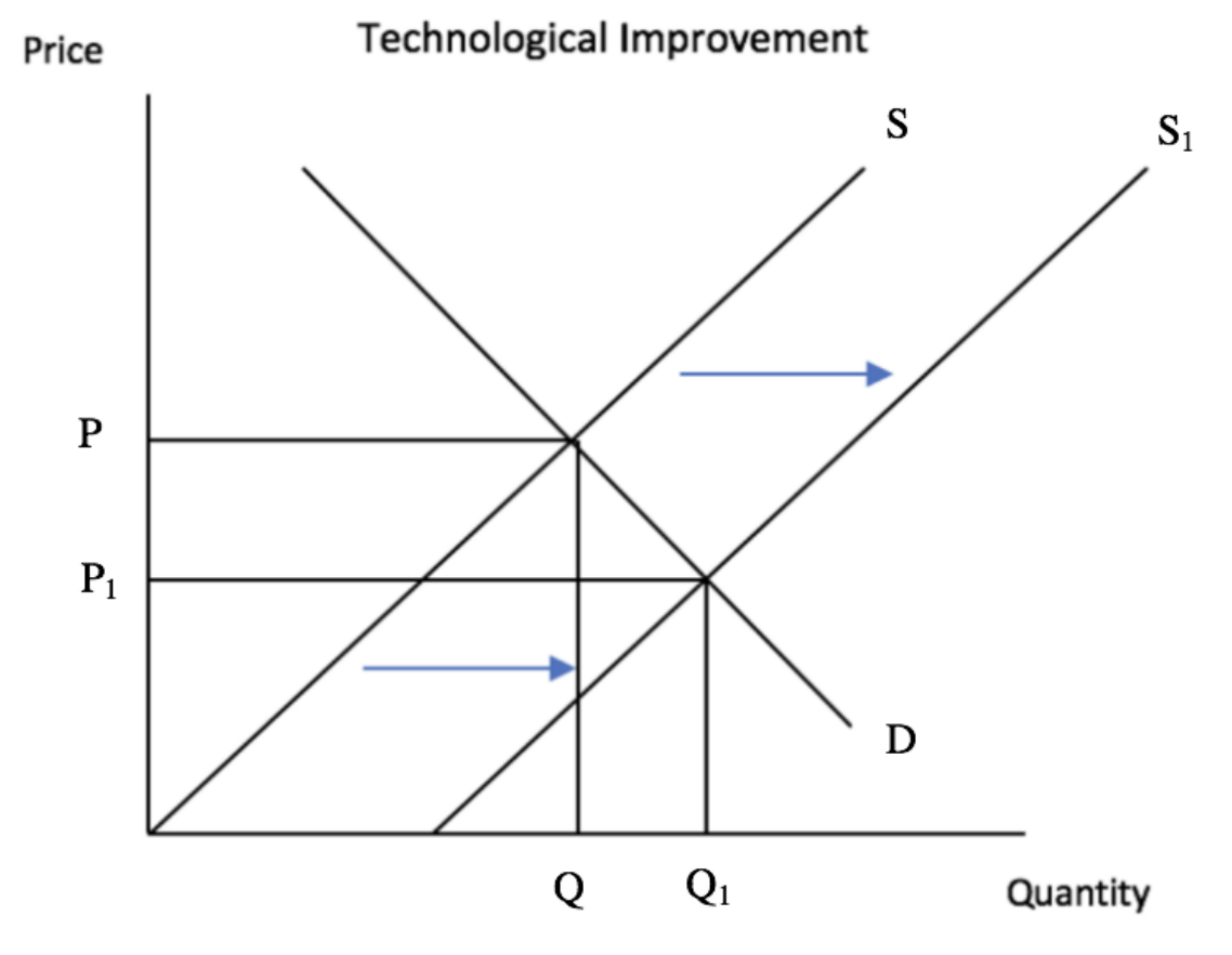


Question 1b

ii) Supply: Technological improvement (with demand fixed)

A technological improvement means more quantity can be produced for the same cost (price). Hence, the supply curve shifts right. Quantity increases to Q_1 and price falls to P_1.

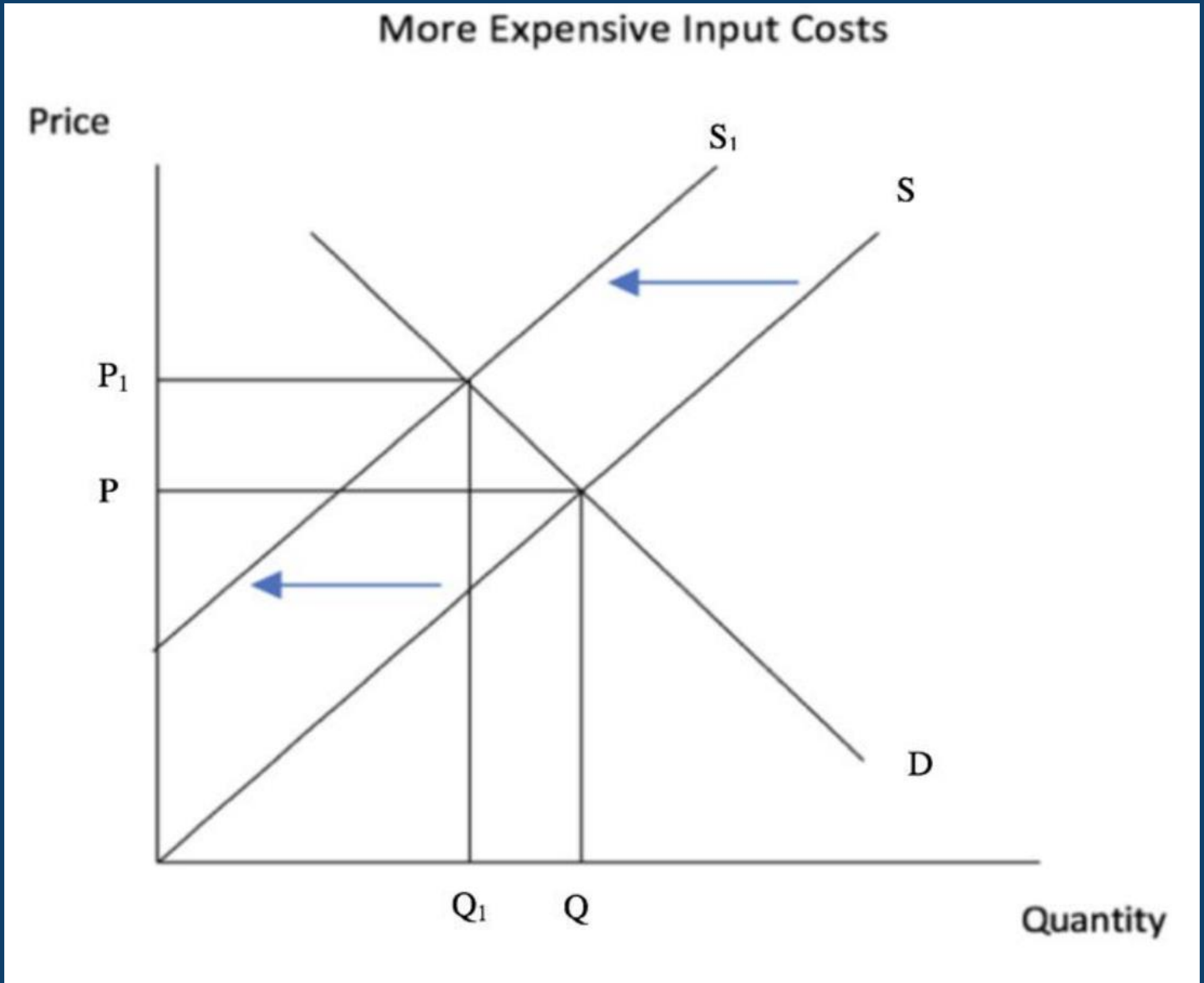
…remember the PPC from a few weeks ago…



Question 1b

iii) Supply: More expensive input costs (with demand fixed)

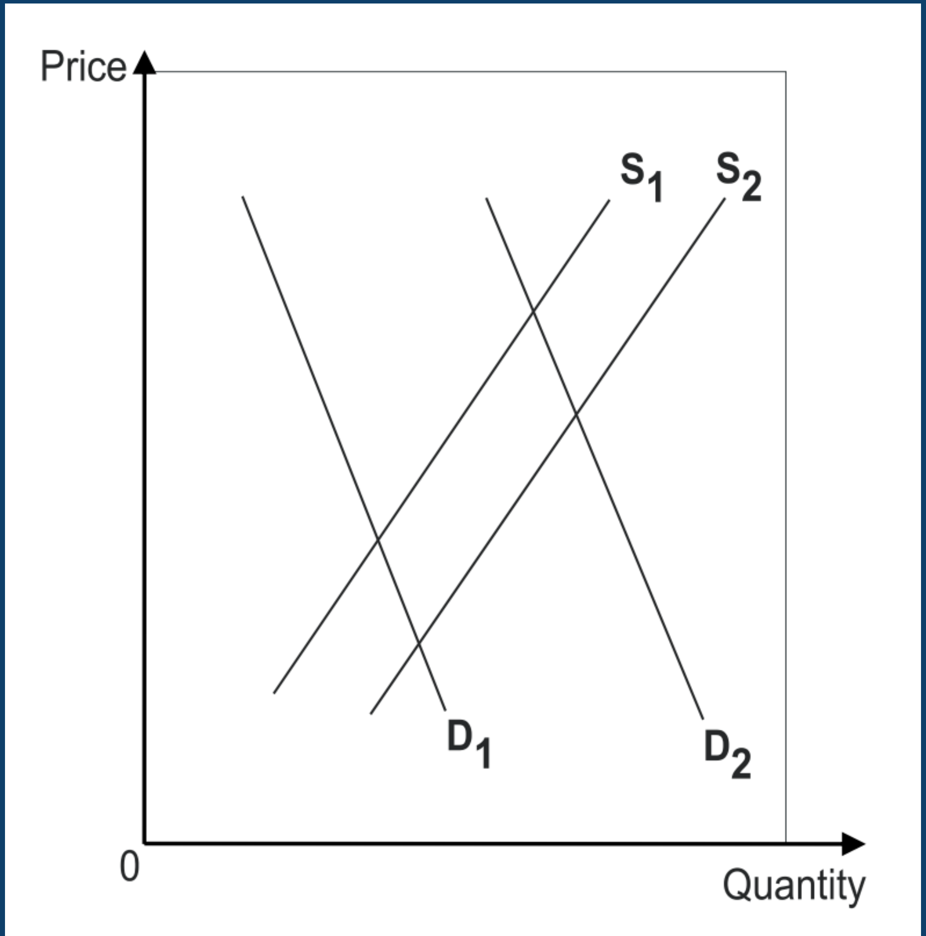
As input costs increase, suppliers will have higher costs for a given quantity produced – resulting in increased prices. The supply curve therefore shifts left. Demand remains unchanged. Price increases to P1 and quantity demanded decreases to Q1.



Question 1c

c) Consider the following diagram. Assume demand remains unchanged at D_1. If supply shifts from S_1 to S_2 then the equilibrium price will **(rise/fall)** and the equilibrium quantity will **(rise/fall/indeterminate)**.

fall; rise



Quick Recap – Price Mechanism

1. Draw and label original d/s with title axes
2. Label equilibrium p and q
3. Ask yourself: does the question effect demand or supply?
4. Is it an increase or decrease in d/s (if non-price factor)?
5. Draw in new curve (and label it)
6. Label new equilibrium p and q
7. Draw in arrows

Final check: labelled curves and axes, title, 2xP 2xQ 2xM.E.

Question 2a

a) In an attempt to promote exercise, the Minister for Health is considering funding an education campaign promoting the benefits of exercise. Consider the UQ gym membership market, use the supply and demand model to show:

i) The impacts on supply and demand

ii) The impacts on price and quantity

iii) Did the policy achieve the intended outcome?

Hint: first think – demand side or supply side? Increase or decrease?

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Complement demand
Tastes

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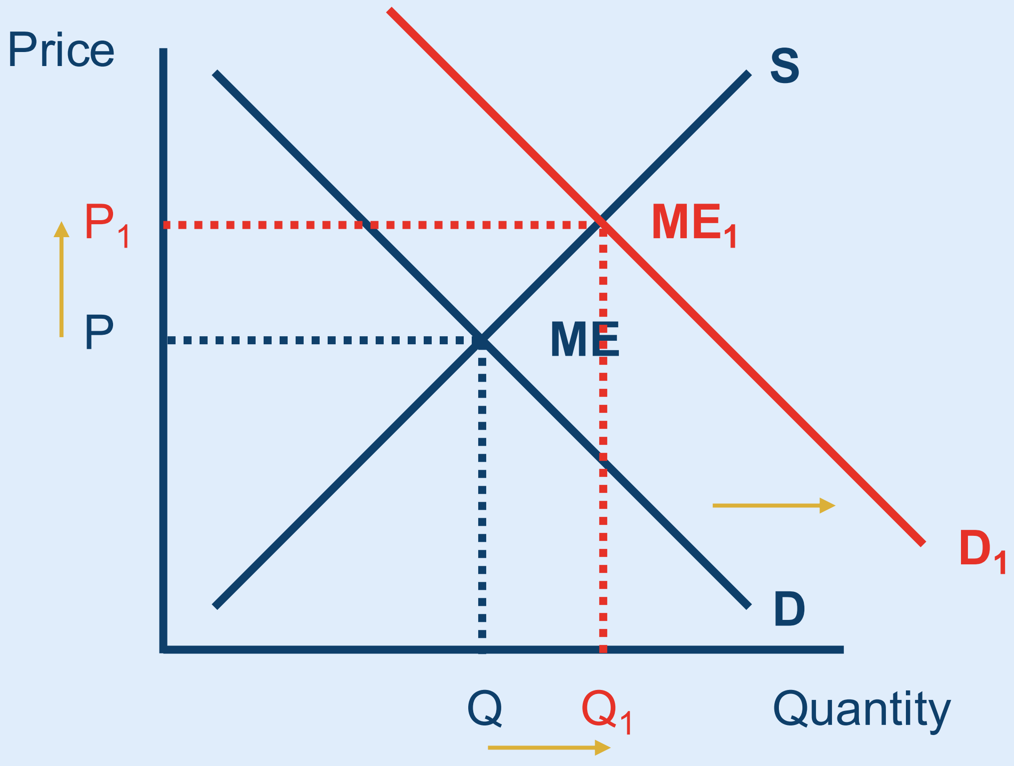
Technology

Cost of making goods
Joint supply

Question 2a

i) The impacts on supply and demand



**The graph shows demand for gym memberships increasing from D to D1 and supply remaining constant.*

ii) The impacts on price and quantity

Price has increased from P to P_1, and quantity demanded/supplied increased to from Q to Q_1.

iii) Did the policy achieve the intended outcome?

As the intended outcome was to increase exercise, the increase in quantity of gym memberships (an indicator of exercise) shows the policy did achieve its goal.

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Question 2a

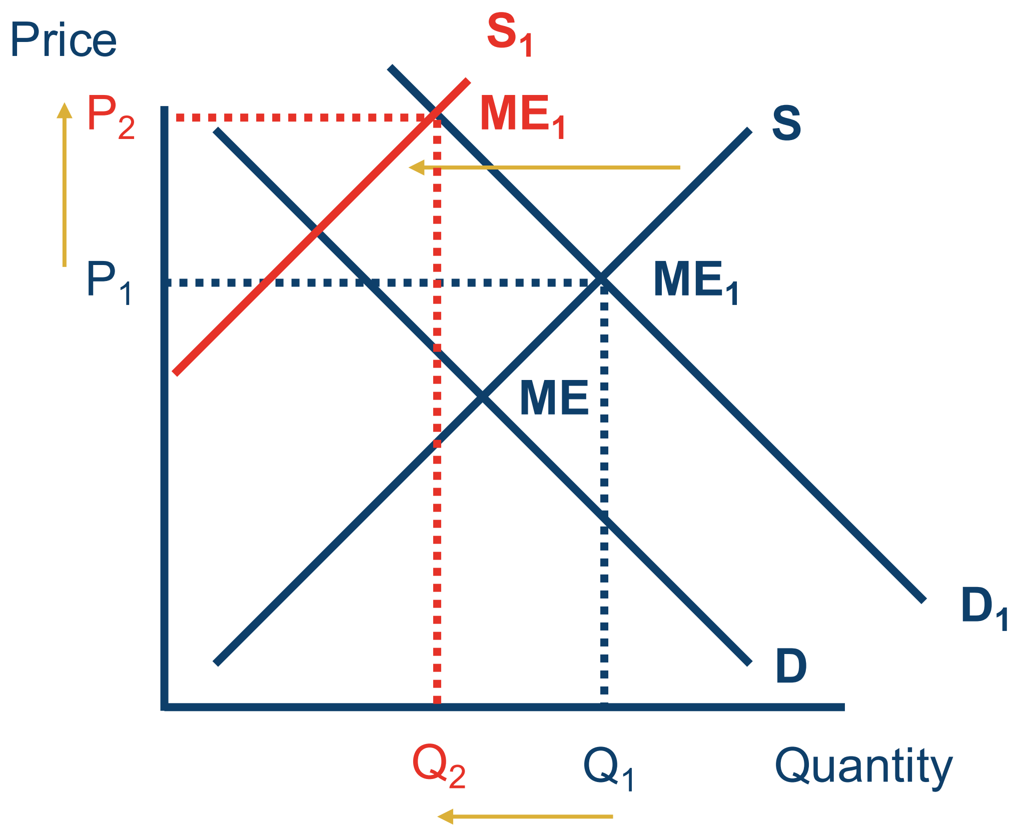
If the UQ gym has to pay higher wages to their current staff and increased electricity rates, show:

iv) The impact on the UQ gym's supply and demand

More expensive input costs result in Supply shifting to the left from S to S_1. Demand remains unchanged at D_1.

v) The impact on price and quantity of memberships

The reduced supply from S to S_1 increases price from P_1 to P_2. Quantity of memberships reduces from Q_1 down to Q_2, but the final impact on quantity demanded is unclear. It will depend on the magnitude of the shift in supply



Question 2b

b) It is known that four wheel drive cars (4WDs) have poor fuel efficiency. Many people consider this issue before purchasing a 4WD.

i) What is the relevant complement good to 4WDs?

ii) Draw one diagram for the supply and demand of 4WDs (with axes labelled) and another diagram for the supply and demand of the complement good (with axes labelled).

iii) How does a negative (leftward) shift in supply of the complement good affect the price and quantity demanded of the complement good (with demand fixed)?

iv) How does the change in the complement good in part iii) affect the demand for 4WDs (with supply fixed)? v)

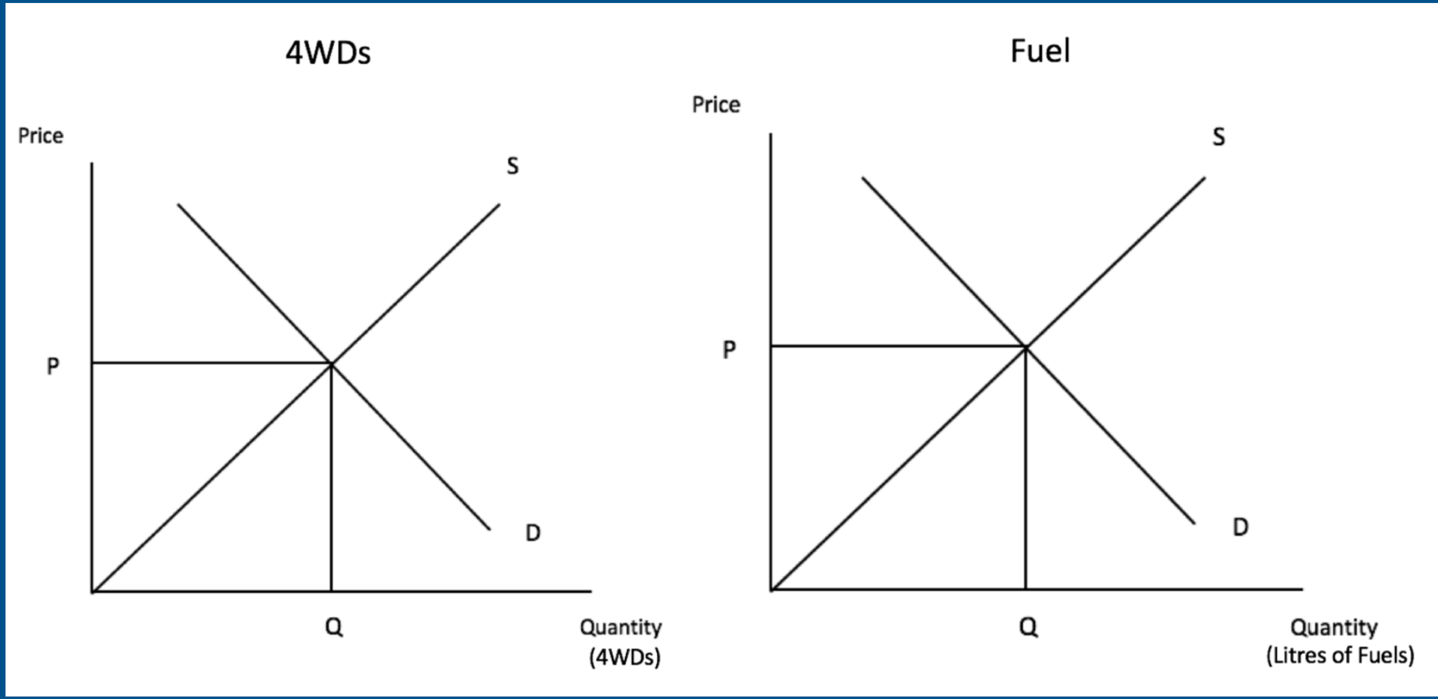
v) How does the change in 4WD demand affect the price and quantity demanded of 4WDs?

Question 2b

i) What is the relevant complement good to 4WDs?

Fuel (willing to accept tyres, although fuel is the most relevant consideration in buying a 4WD)

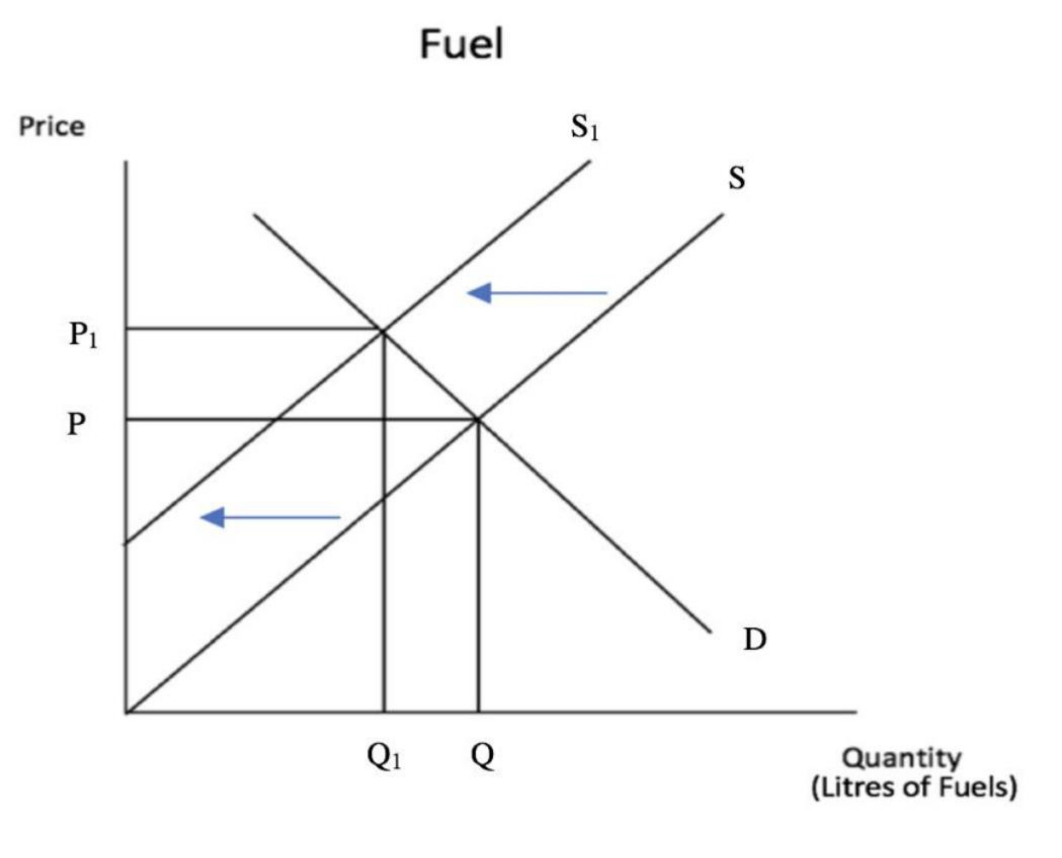
ii) Draw one diagram for the supply and demand of 4WDs (with axes labelled) and another diagram for the supply and demand of the complement good (with axes labelled).



Question 2b

iii) How does a negative (leftward) shift in supply of the complement good affect the price and quantity demanded of the complement good (with demand fixed)?

The leftward shift in supply results in increased prices from P to P1 and decreased quantity from Q to Q1.



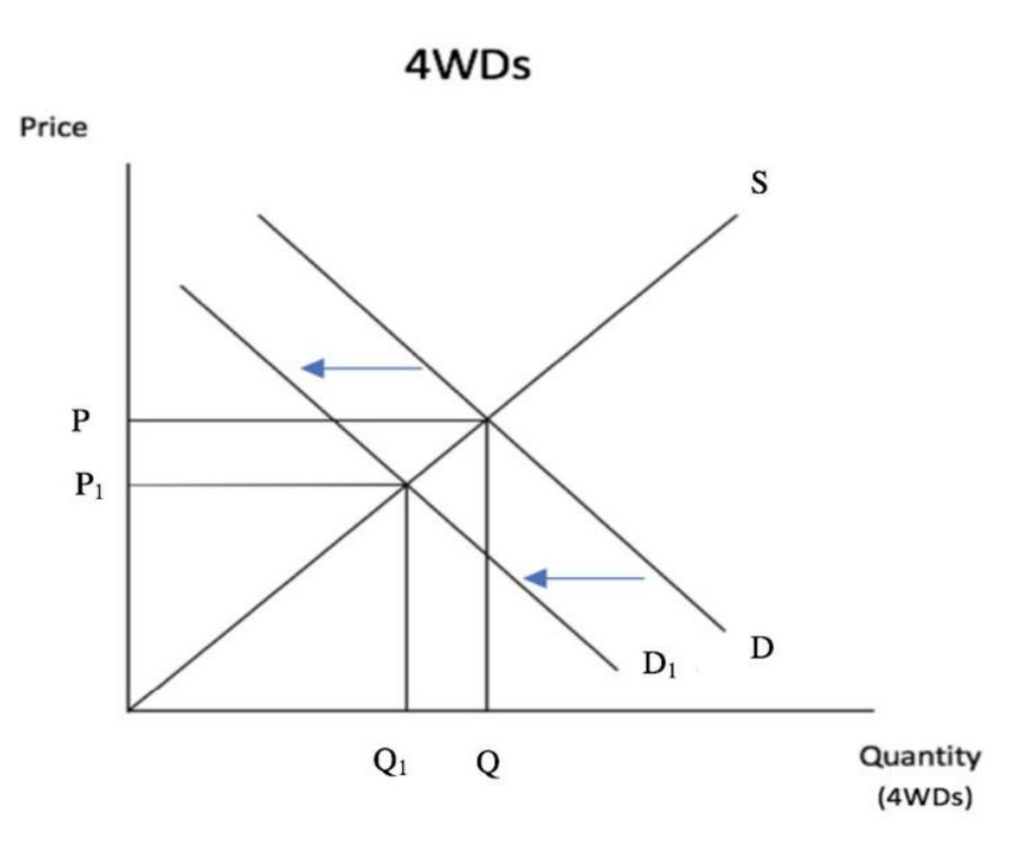
Question 2b

iv) How does the change in the complement good in part iii) affect the demand for 4WDs (with supply fixed)?

As 4WDs require a lot of fuel and less people are now buying fuel, this means less people will buy 4WDs and look for different low fuel consumption vehicles. Therefore, the demand for 4WDs will decrease (leftward shift).

v) How does the change in 4WD demand affect the price and quantity demanded of 4WDs?

The decrease in 4WD demand will decrease prices from P to P_1 and quantity demanded from Q to Q_1.



Question 3

a) Reflect on your own personal situation (example in brackets):

i) Can you and your friends think of any good/service you purchase regularly?

ii) What is a complement good to this good/service? (internet connection)

iii) If this complement good becomes significantly more expensive, would this impact your likelihood to continue paying for the good/service?

iv) What is an example substitute good?

v) How could this substitute good impact your likelihood to use the good/service?

Thank You

Next week: Elasticity

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